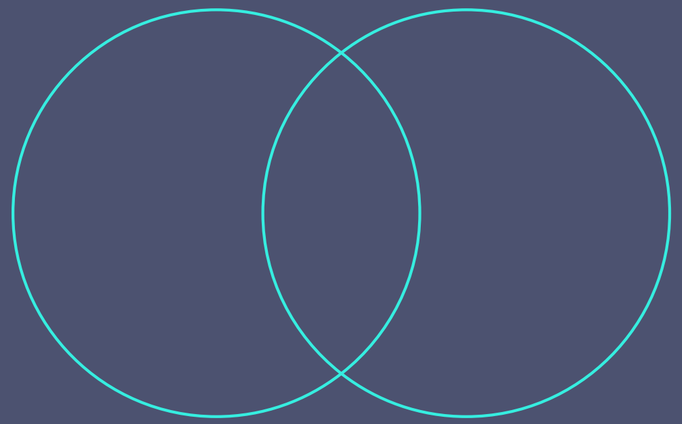
STUDENT HABITS AND ACADEMIC PERFORMANCE EVALUATIO

Data Analytics Capstone Project

1. Problem Statement



Main question:
Which factors are associated with exam performance?

Students have different study habits and academic backgrounds.
Factors such as study hours, sleep, motivation, stress, and previous GPA may be related to exam performance.
This project analyzes the dataset to find important patterns and relationships

2. Objectives

Analyze student habits and exam performance.
Find relationships between student habits and exam scores.
Compare performance across different student groups.
Identify an exploratory student risk profile.
Provide simple, data-driven recommendations.



3. Dataset and Data Preparation

80,000
students

31
columns

0
missing values

0
duplicate records

Checked the structure and data types.
Checked missing values and duplicate records.
Created useful categories such as attendance, sleep, performance, screen time, and stress.

4. APPASA Framework

PREPARE
Understand the data

ASK
Define the problem

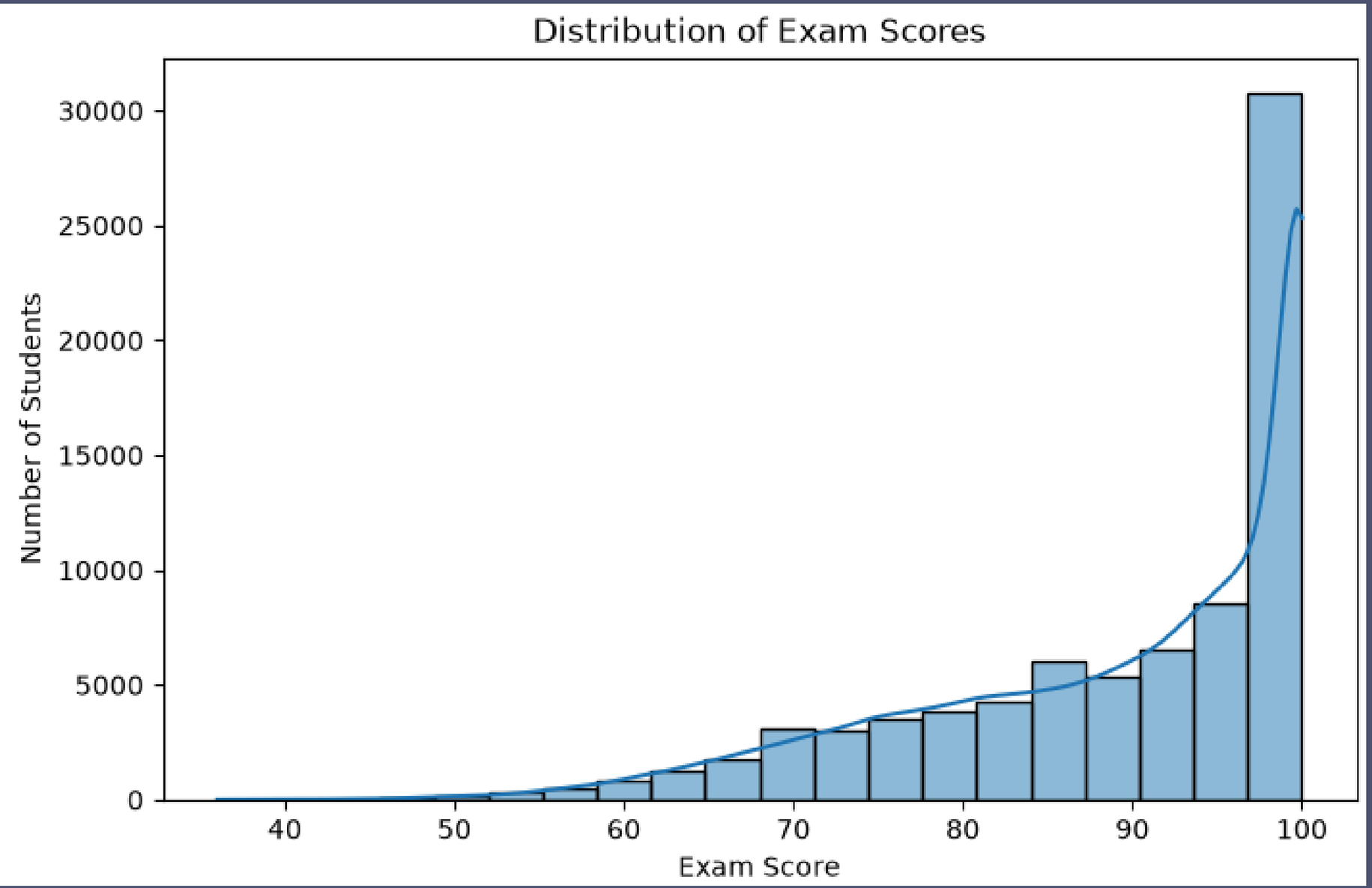
PROCESS
Clean and transform

ANALYZE
Find patterns

SHARE
Present findings

ACT
Give recommendatio

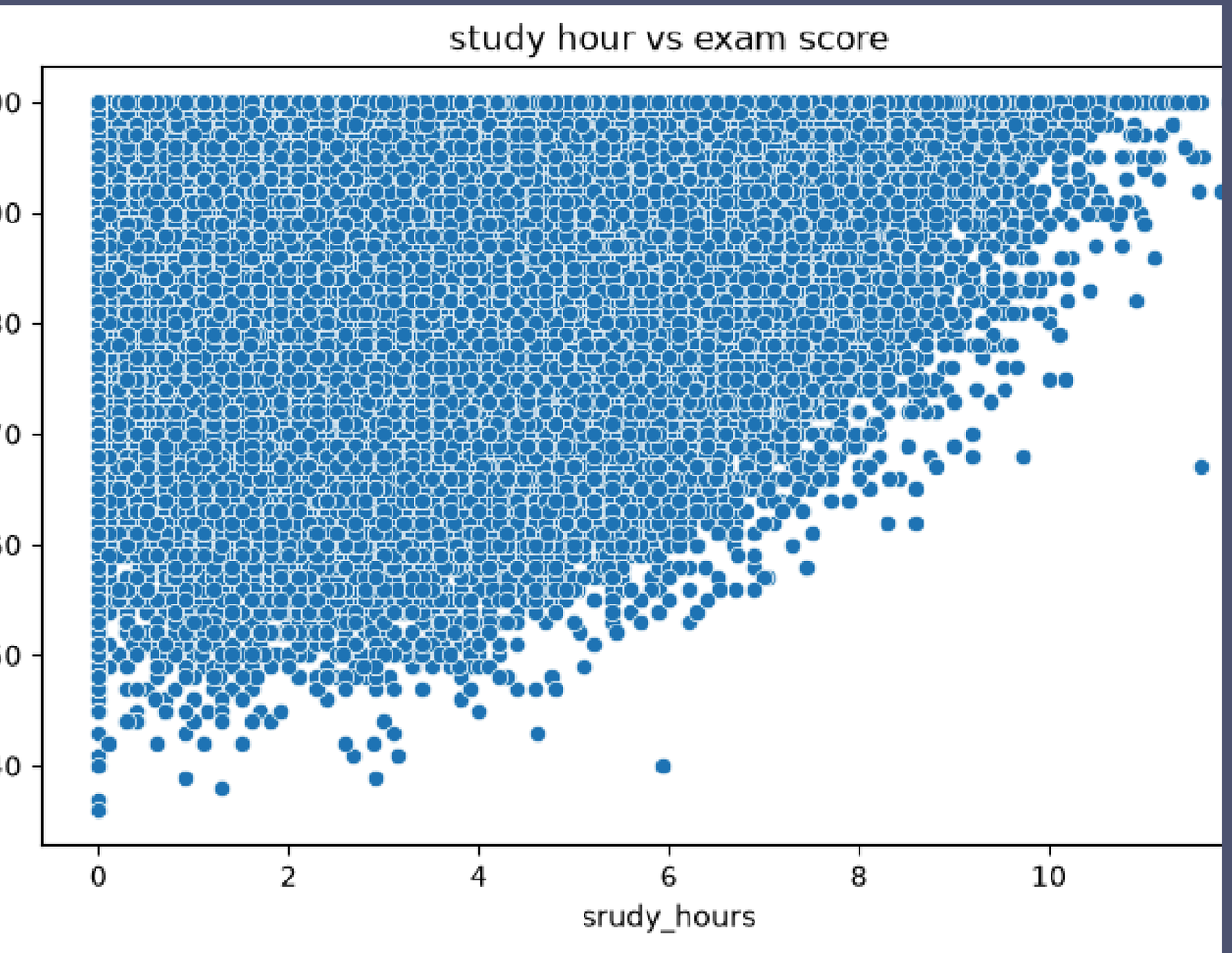
5. Exam Performance

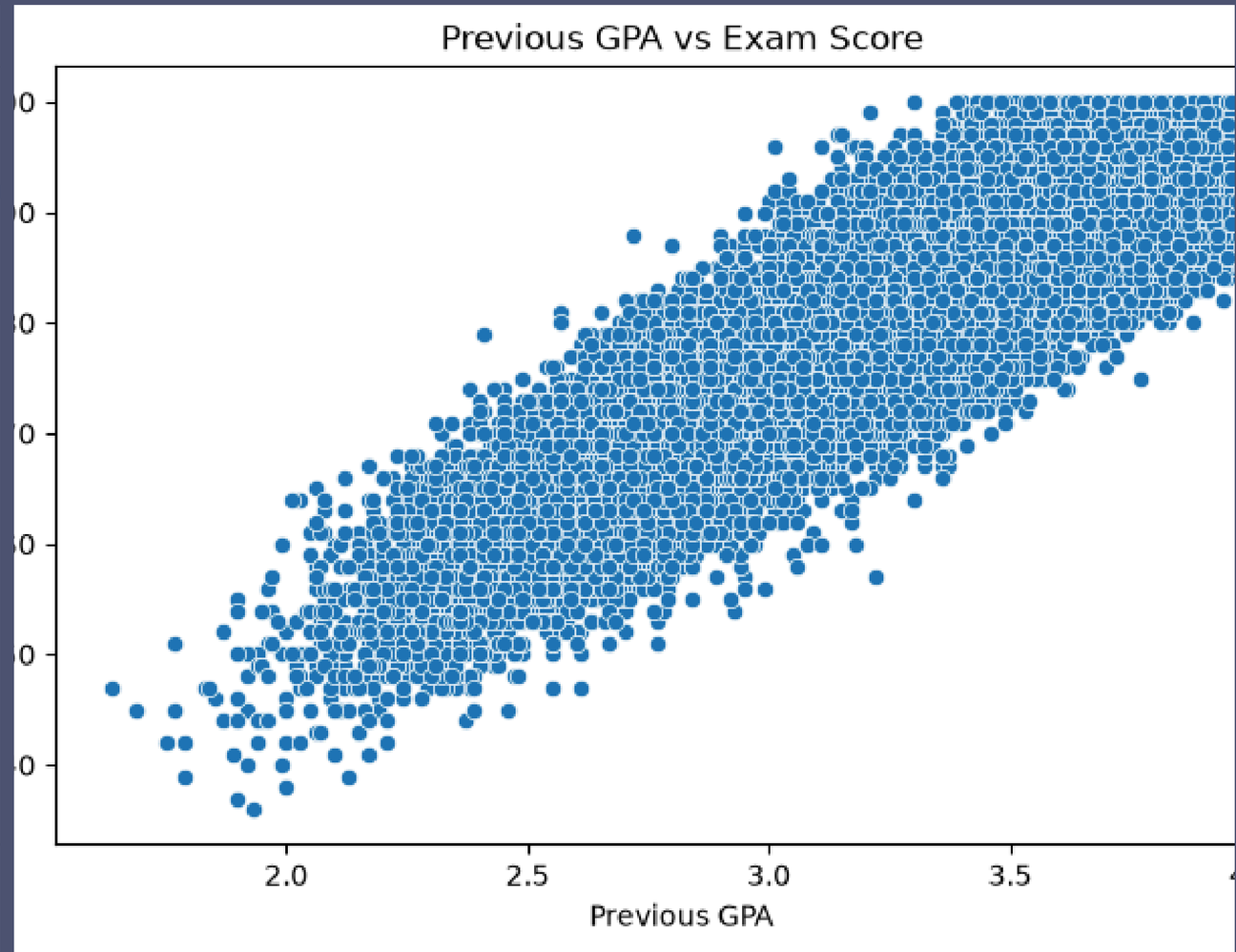


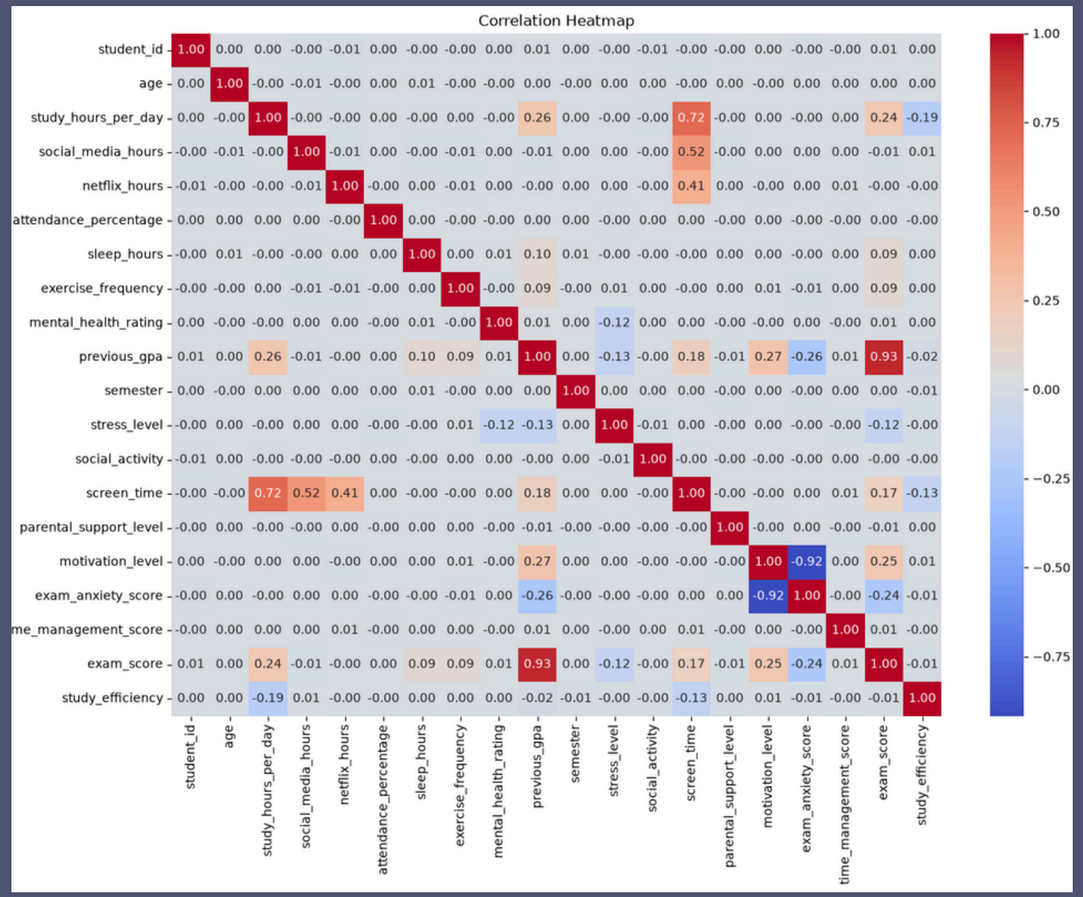
Average exam score ≈ 89.14

Most scores are in the higher range.

6. Important Relationships







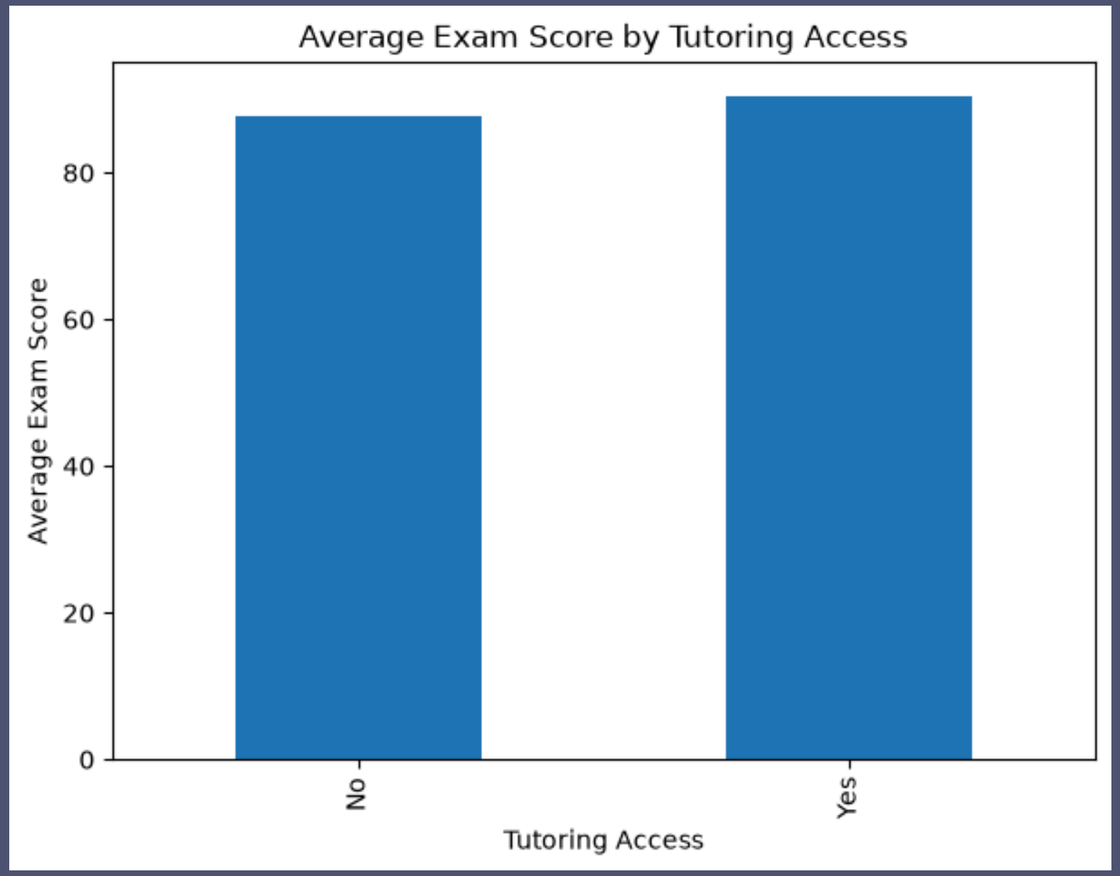
7. Correlation Analysis

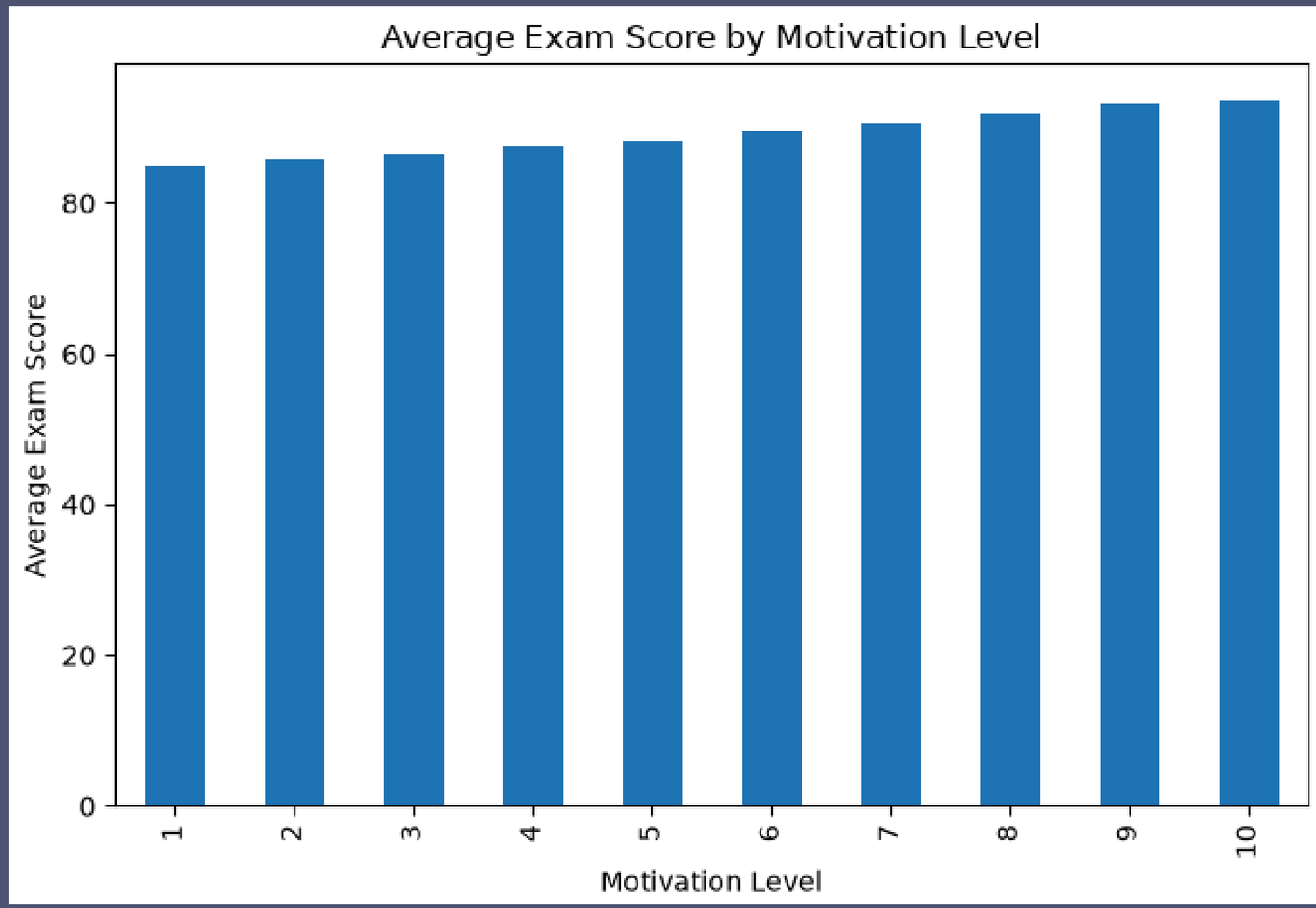
Previous GPA
r ≈ 0.93

Motivation
r ≈ 0.25

Study Hours
r ≈ 0.24

8. Group Analysis





9. Key Findings

Previous GPA had the strongest positive relationship with exam score (r ≈ 0.93).
Study hours and motivation showed positive relationships with exam performance.
Exam anxiety and stress showed negative relationships with exam score.
Students with tutoring had a higher average score in this dataset.
Attendance and sleep showed weaker relationships with exam performance.

10. Conclusion and Recommendations

Conclusion

Previous GPA showed the strongest relationship with exam score.
Study hours and motivation showed positive relationships.
Exam anxiety and stress showed negative relationships.
These findings describe patterns and do not prove causation.

Recommendations

Study consistently and manage study time.
Maintain motivation and clear academic goals.
Manage exam anxiety and stress.
Get sufficient and regular sleep.
Seek tutoring or academic support when needed.